

Exam: Correlation (Pearson, Spearman)

Multiple Choice (Single Select) - 4 Answer Options

1. What does Pearson's correlation coefficient measure?

- A) The variance explained
- B) The causal effect
- C) The linear relationship between two metric variables
- D) The relationship between two rank orders

2. What is the range of r ?

- A) -infinity to +infinity
- B) -1 to +1
- C) 0 to 1
- D) 0 to 100

3. An r of -0.8 indicates...

- A) A strong negative relationship
- B) A strong positive relationship
- C) A weak negative relationship
- D) No relationship

4. When do you use Spearman's rho?

- A) Only with nominal scales
- B) With ordinal data or when linearity is violated
- C) Only with normally distributed data
- D) Only with interval data

5. What is the squared correlation coefficient?

- A) The variance
- B) The coefficient of determination (r^2)
- C) The covariance
- D) The effect size

6. What does correlation NOT imply?

- A) A statistical relationship
- B) A linear relationship
- C) Causation
- D) An association

7. Which outlier can strongly influence r ?

- A) Any outlier
- B) Only outliers in Y
- C) Only outliers in X
- D) Influential point (leverage point)

8. Which correlation measures rank-order relationships?

- A) Pearson's r
- B) Spearman's ρ
- C) Pearson's chi-square
- D) Cramer's V

Solutions

1: C

2: B

3: A

4: B

5: B

6: C

7: D

8: B